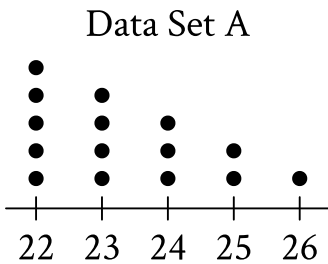


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question



The dot plot represents the **15** values in data set A. Data set B is created by adding **56** to each of the values in data set A. Which of the following correctly compares the medians and the ranges of data sets A and B?

Answer

- A. The median of data set B is equal to the median of data set A, and the range of data set B is equal to the range of data set A.
- B. The median of data set B is equal to the median of data set A, and the range of data set B is greater than the range of data set A.
- C. The median of data set B is greater than the median of data set A, and the range of data set B is equal to the range of data set A.
- D. The median of data set B is greater than the median of data set A, and the range of data set B is greater than the range of data set A.

Correct Answer: C

Rationale

Choice C is correct. The median of a data set with an odd number of values, in ascending or descending order, is the middle value of the data set, and the range of a data set is the positive difference between the maximum and minimum values in the data set. Since the dot plot shown gives the values in data set A in ascending order and there are **15** values in the data set, the eighth value in data set A, **23**, is the median. The maximum value in data set A is **26** and the minimum value is **22**, so the range of data set A is **26 — 22**, or **4**. It's given that data set B is created by adding **56** to each of the values in data set A. Increasing each of the **15** values in data set A by **56** will also increase its median value by **56** making the median of data set B **79**. Increasing each value of data set A by **56** does not change the range, since the maximum value of data set B is **26 + 56**, or **82**, and the minimum value is **22 + 56**, or **78**, making the range of data set B **82 — 78**, or **4**. Therefore, the median of data set B is greater than the median of data set A, and the range of data set B is equal to the range of data set A.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.